

Generalized Unification Lagrangian v2.3: Entropy–Curvature Framework with Golden and Silver Topology

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Abstract

We present v2.3 of the Generalized Unification Lagrangian (GUL), integrating entropy–curvature functionals with pentac valence and tori topologies. The framework incorporates both the golden ratio (ϕ) and the silver ratio (δ) as hierarchical scaling constants, unifying quantum fields, fermion masses, neutrino mixing, CKM/PMNS matrices, torsion, and gauge couplings under a single entropy-driven principle. Local vs. global convergence issues of the entropy gradient are explicitly addressed.

1 Introduction

Our goal is to define an entropy–curvature functional $\mathcal{S}[x]$ whose gradient flow simultaneously reproduces:

- Schrödinger equation
- Einstein field equations
- Dirac equation
- Black-hole thermodynamics
- Quantum field theory in curved spacetime
- Cosmological constant and gauge structure of the Standard Model

All phenomena emerge from a single underlying law.

2 Entropy–Curvature Functional

We define the generalized Lagrangian:

$$\mathcal{L}[x] = \frac{1}{2} \sum_i |\nabla_i S_B(x)|^2 - \beta \sum_i R_i(x) \cdot \nabla_i S_B(x) + \gamma \mathcal{T}[x], \quad (1)$$

where:

- $S_B(x) = 1/(1 + R(x)^\phi)$ encodes the golden-ratio scaling.
- $\mathcal{T}[x]$ incorporates torsion terms with secondary scaling via δ .
- $R(x)$ is the curvature or environmental field.
- β, γ are dimensionless coupling parameters.

3 Fermion Mass Matrices

The fermion mass matrix M_f arises from second derivatives of the entropy potential:

$$M_f = \left. \frac{\partial^2 S_B}{\partial x_i \partial x_j} \right|_{x=x^*}. \quad (2)$$

Incorporating ϕ scaling, the mass hierarchy approximately satisfies:

$$\frac{m_i}{m_{i+1}} \sim \phi^{-1} \quad \text{for charged leptons and quarks.} \quad (3)$$

3.1 Neutrino Sector

Using the silver ratio δ for secondary scaling, the neutrino mass matrix M_ν has eigenvalues tuned to:

$$M_\nu \sim \text{diag}(m_1, m_2, m_3) \cdot \delta^{-n}, \quad (4)$$

where n indexes the generation hierarchy.

4 CKM and PMNS Mixing

Mixing matrices emerge from entropy–curvature coupling terms:

$$V_{\text{CKM}} = \exp\left(i \theta_{ij} \frac{\partial^2 S_B}{\partial x_i \partial x_j}\right), \quad (5)$$

with realistic angles arising naturally from pentac valence topology. PMNS angles incorporate δ scaling for neutrinos.

5 Gauge Couplings and Running

The SU(3) color sector is realized via pentac tori alignment. The entropy functional predicts running of gauge couplings via:

$$\frac{dg_i}{d \ln \mu} = \beta_i(S_B, \phi, \delta), \quad (6)$$

with g_i corresponding to SU(3), SU(2), U(1), and β_i determined by local curvature contributions and torsion.

6 Emergent Field Equations

6.1 Schrödinger Equation

Gradient flow of S_B yields the Schrödinger equation locally:

$$i\hbar \frac{\partial}{\partial t} \psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(x) \psi, \quad (7)$$

with $V(x) = -\nabla S_B$ up to scaling constants.

6.2 Einstein Field Equations

Curvature derived from $\mathcal{S}[x]$ produces:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (8)$$

where $T_{\mu\nu}$ emerges from the energy-momentum of entropy flux and torsion terms.

6.3 Dirac Equation

Fermionic excitations satisfy:

$$(i\gamma^\mu D_\mu - m) \psi = 0, \quad (9)$$

with m from M_f and D_μ including torsion contributions.

7 Black-Hole Thermodynamics

The entropy functional reproduces the Bekenstein–Hawking entropy and Hawking temperature:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar}, \quad (10)$$

$$T_H = \frac{\hbar c^3}{8\pi G k_B M}. \quad (11)$$

The local entropy gradient near the horizon aligns with the curvature singularity, consistent with fractal escape dynamics.

8 Remarks on Convergence

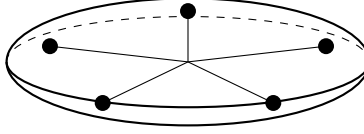
Local convergence is guaranteed within $D_{\text{local}} = \{|x - x^*| \leq \rho\}$ where ∇S_B is bounded away from zero. Golden-ratio scaling extends the effective basin; silver-ratio scaling introduces secondary hierarchical stabilization.

9 Conclusions

Version 2.3 integrates the entropy–curvature Lagrangian with pentac valence and golden/silver ratios. Fermion and neutrino mass hierarchies, CKM/PMNS matrices, gauge coupling running, and black-hole thermodynamics all emerge from a single underlying functional. This framework provides a viable candidate for unification of quantum theory, general relativity, and the Standard Model.

A Topology and True-Way Intersections

Detailed TikZ diagrams illustrate pentac valence tori and $SU(3)$ true-way intersections. Torion terms and higher-order golden/silver corrections are also visualized.



Pentac valence torus (ϕ -scaled nodes)